MTA Daily Ridership – Entity Relationship Diagram (Star Schema)

DEPI · Data Analysis Track · Phase 2 · Design & Analysis · v1 – field names match the wireframes

Dim_Date
dimension · 1,706 rows

PK	**Date**	DATE	2024-10-31
	Year	INT	2024
	MonthNo	INT	10
	Month	TEXT	Oct
	MonthStart	DATE	2024-10-01
	YearMonth	TEXT	2024-10
	WeekdayNo	INT	4
	Weekday	TEXT	Thu
	DayType	TEXT	Weekday
	Period	TEXT	Current

Created in DAX (CALENDAR 1 Mar 2020 – 31 Oct 2024) and marked as the date table

1 — Date — *

Fact_Ridership
fact · 11,942 rows (1,706 days × 7 modes)

PK, FK	**Date**	DATE	→ Dim_Date
PK, FK	**Mode**	TEXT	→ Dim_Mode
	Ridership	INT	daily count
	PctPrePandemic	DECIMAL	75 = 75%

* — Mode — 1

Dim_Mode
dimension · 7 rows

PK	**Mode**	TEXT	Subways
	Unit	TEXT	Riders
	ModeGroup	TEXT	Passenger

Built from the distinct Mode values (Reference + Remove Duplicates)
Unit and ModeGroup added with a conditional column

Unpivot + clean in Power Query (Date + Mode + Ridership + PctPrePandemic)

Raw_MTA
staging · 1,706 rows · 15 columns · not loaded

Date	DATE	1 column
<Mode>: Total …	INT	7 columns
<Mode>: % of …	INT	7 columns

Legend

PK Primary key (Fact: Date + Mode together)

FK Foreign key to a dimension

One-to-many: one dimension row, many fact rows

Relationships run one direction (Dimension → Fact) in Power BI.

Dashed box = staging only, not loaded in the model.

Modeling rules

- Total Passenger Ridership uses ModeGroup = Passenger: Subways, Buses, LIRR, Metro-North, Staten Island Railway.
- Access-A-Ride (scheduled trips) and Bridges and Tunnels (vehicle traffic) stay in the model but are never added to passenger totals.
- Staten Island zero values are reviewed in Power Query and converted to null where they are missing observations.
- PctPrePandemic is stored as percent points (75 = 75%), so the 100% reference line is at 100.
- Dim_Date is marked as the date table; Month sorts by MonthNo and Weekday by WeekdayNo.